# Rates of Change and Rectilinear Motion



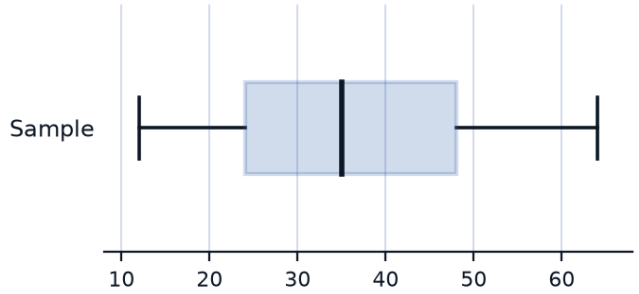
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

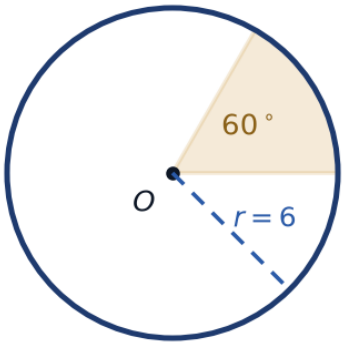
**2** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A continuous random variable $X$ has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant $k$; (b) the median of $X$; (c) the variance $\mathrm{Var}(X)$.




**3** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

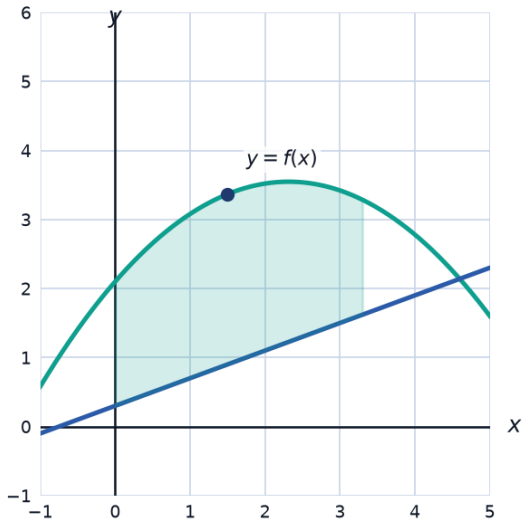
Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \le \pi$. Hence find $z^6$ and determine all three solutions to $w^3 = z$.




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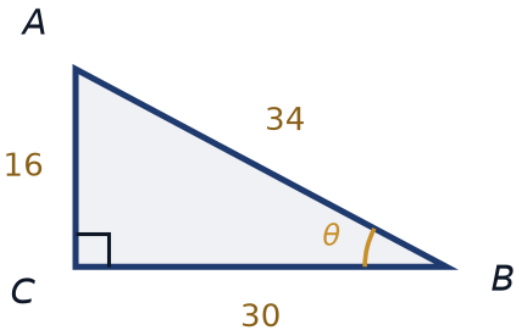
**4** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



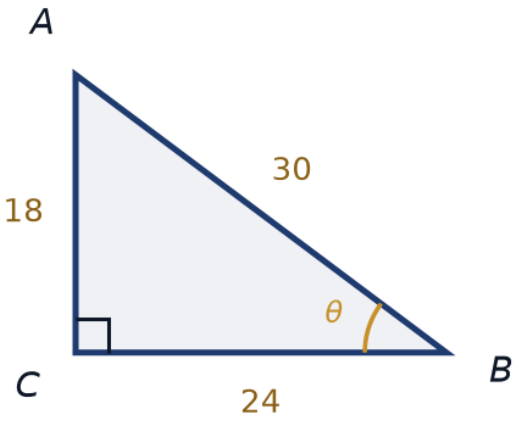

**5** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Give the first three nonzero terms of the Maclaurin series for $\sin x$.




**6** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Give the first three terms of the Maclaurin series for $\cos x$.




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**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that $L_1$ and $L_2$ are skew, and find the shortest perpendicular distance between them.

**9** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For what $x$ does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

**10** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x \, dx$ for integers $n \geq 0$. (a) Evaluate $I_0$. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x \, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 2 · MediumFind the point $(x, y)$ on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 3 · Advanced / AP FRQEvaluate the improper integral $\int_0^\infty xe^{-2x}\, dx$ or establish that it diverges.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle moves along the $x$-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \ge 0$. At what times is the particle at rest?

**a.** $t = 1$ and $t = 3$

**b.** $t = 0$ and $t = 3$

**c.** $t = 2$

**d.** $t = 3$ only

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**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

**a.** $8$

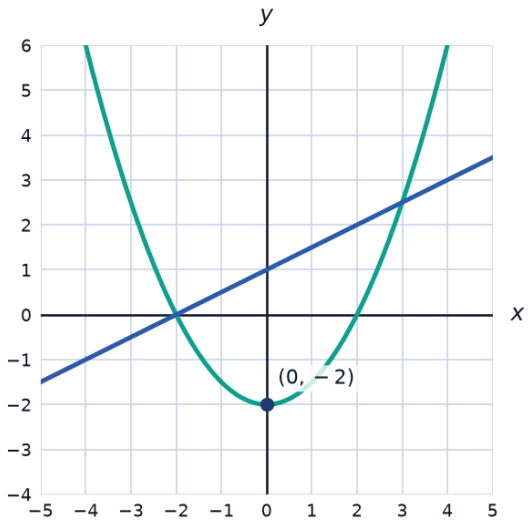
**b.** $-8$

**c.** $5$

**d.** $\sqrt{73}$

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When is a particle's speed increasing?




**a.** When velocity and acceleration have the same sign

**b.** When velocity is positive

**c.** When acceleration is positive

**d.** When acceleration is zero

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

**a.** $-3$

**b.** $5$

**c.** $-4$

**d.** $3$

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**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

**a.** 5

**b.** 3

**c.** 4

**d.** 7

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

**a.** $\int_a^b |v(t)|dt$

**b.** $\int_a^b v(t)dt$

**c.** $|\int_a^b v(t)dt|$

**d.** $\int_a^b a(t)dt$

**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

**a.** $60\pi$ cm$^2$/s

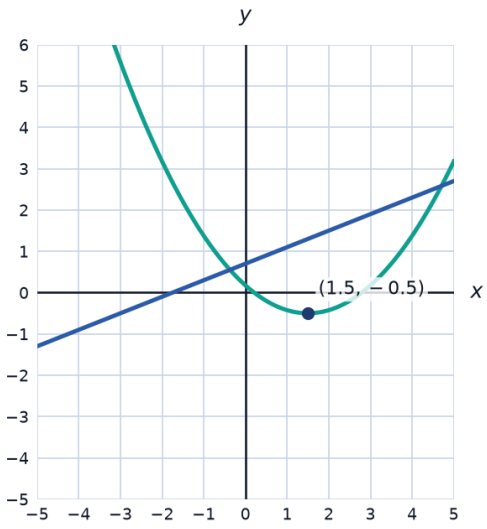
**b.** $30\pi$ cm$^2$/s

**c.** $100\pi$ cm$^2$/s

**d.** $20\pi$ cm$^2$/s

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?




**a.** $-\frac{5}{6}$ ft/s

**b.** $-\frac{12}{5}$ ft/s

**c.** $-2$ ft/s

**d.** $-\frac{5}{12}$ ft/s

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Air is pumped into a spherical balloon at a rate of $100$ cm$^3$/s. How fast is the radius increasing when $r = 5$ cm?

**a.** $\frac{1}{\pi}$ cm/s

**b.** $\frac{1}{4\pi}$ cm/s

**c.** $\frac{4}{\pi}$ cm/s

**d.** $25\pi$ cm/s

**30** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A conical tank with vertex down has height $10$ m and base radius $4$ m. Water flows in at $2$ m$^3$/min. What is the relation between $r$ and $h$?

**a.** $r = \frac{2}{5}h$

**b.** $r = \frac{5}{2}h$

**c.** $r = \frac{1}{2}h$

**d.** $r = 4h$

**31** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m$^3$/min?

**a.** $\frac{1}{2\pi}$ m/min

**b.** $\frac{2}{\pi}$ m/min

**c.** $\frac{1}{4\pi}$ m/min

**d.** $\frac{4}{25\pi}$ m/min

**32** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A street light is mounted at the top of a $15$-foot pole. A $6$-foot man walks away from the pole at $5$ ft/s. How fast is the tip of his shadow moving?

**a.** $\frac{25}{3}$ ft/s

**b.** $5$ ft/s

**c.** $\frac{10}{3}$ ft/s

**d.** $10$ ft/s

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**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region $S$ in the $uv$-plane with area 2 maps to a region $R$. Find the area of $R$.

**a.** 2

**b.** 8

**c.** 12

**d.** 16

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

**a.** $40\pi$ cm³/s

**b.** $100\pi$ cm³/s

**c.** $200\pi$ cm³/s

**d.** $300\pi$ cm³/s

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Related rates: ladder 10 m against wall. Base slides at 1 m/s when base is 6 m. Rate of top:

**a.** $-3/4$ m/s

**b.** $3/4$ m/s

**c.** $-1$ m/s

**d.** $-4/3$ m/s

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Change order: $\int_0^2 \int_0^y f\,dx\,dy$ becomes $\int_0^2 \int_?^? f\,dy\,dx$:

**a.** $\int_0^2 \int_x^1 f\,dy\,dx$

**b.** $\int_0^2 \int_0^x f\,dy\,dx$

**c.** $\int_0^1 \int_x^2 f\,dy\,dx$

**d.** $\int_0^2 \int_x^2 f\,dy\,dx$

**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The change-of-variables formula $\iint_D f\,dA = \iint_{D^*} f(g)\,|J|\,du\,dv$ requires $Je0$ (injective map). The factor $|J|$ is the:

**a.** Laplacian

**b.** Hessian

**c.** Jacobian determinant

**d.** divergence

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Change of variables: $\iint_D f(x,y)\,dA = \iint_{D'} f(x(u,v),y(u,v))|J|\,du\,dv$ where $J =$

**a.** $x_u y_v + x_v y_u$

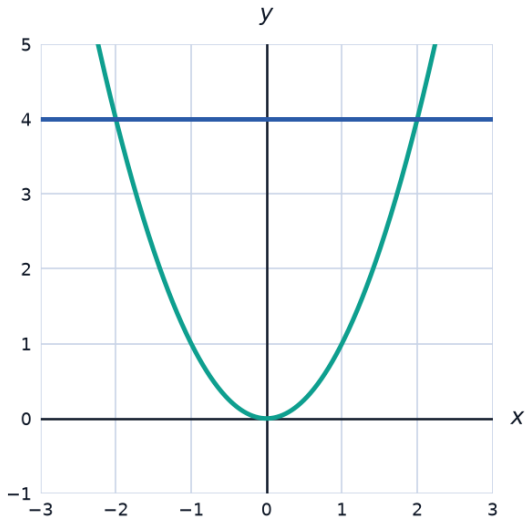
**b.** $x_u y_v - x_v y_u$

**c.** $x_u + y_v$

**d.** $1$

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Consider the region $R$ in the $xy$-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?




**a.** $\frac{32}{3}$

**b.** $\frac{16}{3}$

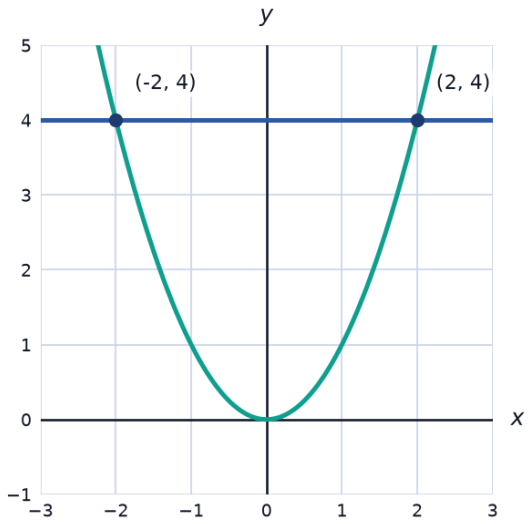
**c.** $8$

**d.** $16$

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**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The figure shows the region $R$ bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region $R$ using the order of integration $dydx$?




**a.** $\int_{-2}^{2}\int_{x^2}^{4} dydx$

**b.** $\int_{-2}^{2}\int_{4}^{x^2} dydx$

**c.** $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

**d.** $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area Between Curves

A particle's motion is described by the parametric equations $x(t) = e^{2t} - 2t$ and $y(t) = 4e^t$ for $t \geq 0$. Find the value of $\frac{dy}{dx}$ at the point where the tangent line to the curve is horizontal.

**a.** 0

**b.** 1

**c.** 2

**d.** 4

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**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Consider the solid $E$ bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of $\pi$.

**a.** $\frac{16\pi}{3}$

**b.** $\frac{32\pi}{3}$

**c.** $16\pi$

**d.** $\frac{64\pi}{3}$

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Limits Graphically and Numerically

The rate of change of a quantity $Q(t)$ is given by $\frac{dQ}{dt} = kte^{-0.5t^2}$, where $k$ is a positive constant. If $Q(0) = 5$ and $Q(2) = 15$, find the value of $k$.

**a.** $\frac{5}{1 - e^{-2}}$

**b.** $\frac{5}{e^{-2} - 1}$

**c.** $\frac{10}{1 - e^{-2}}$

**d.** $\frac{10}{e^{-2} - 1}$

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time $t$ (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

**a.** 210

**b.** 208

**c.** 212

**d.** 209

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**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 The Definition of the Derivative

The rate of change of the volume of a spherical balloon being inflated is given by $\frac{dV}{dt} = 12\pi \text{ cm}^3/\text{s}$. If the initial volume of the balloon is $36\pi \text{ cm}^3$, find the volume of the balloon after $t$ seconds. The volume $V$ of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.

**a.** $V(t) = 12\pi t + 36\pi$

**b.** $V(t) = \frac{4}{3}\pi t^3 + 36\pi$

**c.** $V(t) = 4\pi t^3 + 36\pi$

**d.** $V(t) = 12\pi t^3 + 36\pi$

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